

scCCVGBen — a single-cell centroid-coupled VAE benchmark

Zeyu Fu and co-authors · School of Computer Science · April 29, 2026

Dear Editor,

We submit a revised version of the manuscript previously titled “CCVGAE: a centroid-coupled variational graph attention autoencoder for stable and interpretable single-cell representation learning” for further consideration.

*The revision reflects a title change to **scCCVGBen** — a single-cell centroid-coupled VAE benchmark. The rename addresses three structural changes prompted by reviewer feedback. The dataset cohort grew from 170 to 200 curated profiles (100 scRNA-seq, 100 scATAC-seq), giving the comparison axes the statistical room they require. The encoder backbone, graph construction, dataset, and evaluation suite were separated into independent axes so that sensitivity to each factor can be read off independently. Figure provenance was locked to the released benchmark manifest so that every panel can be regenerated end-to-end without re-running training. The algorithmic contribution is unchanged, but the manuscript now reads as a benchmark companion rather than a method-introduction paper.*

The principal architectural and experimental changes are as follows:

- ▶ 200-dataset cohort with a full provenance manifest and public-safe supplementary table.
- ▶ Encoder-axis robustness across 14 backbones (Fig. 5).
- ▶ Graph-construction-axis robustness across 5 similarity definitions (Fig. 6).
- ▶ Cross-method scRNA benchmark with 12 completed external baselines (Fig. 7).
- ▶ Cross-method scATAC benchmark with 3 modality-specific baselines (Fig. 8).
- ▶ Three biological case studies with paired conditions (Figs. 9–11).
- ▶ Completed clustering-method sensitivity and seed-control clarifications addressing R1.2, R1.3 and R2.6 without relying on unreported entries.
- ▶ Hyperparameter-default justification and a documented sensitivity grid addressing R2.2 while keeping claims limited to completed benchmark results.
- ▶ Discussion of current foundation-model comparators, including scGPT and scFoundation, without adding unverified figure columns.
- ▶ Per-dataset summary table (Supplementary Table S1) built from the released benchmark manifest — addresses R1.1.
- ▶ Loss equation correction — the adjacency-reconstruction term $w_{\text{adj}} \cdot \|\hat{A} - A\|_F^2$ is now explicit in Equation 11 (it appeared in the architecture caption and Table 1 but was absent from the equation), addressing R2.1.

Two point-by-point rebuttal letters follow this cover letter, one per reviewer.

We thank the reviewers for their constructive comments, which materially improved the scope and presentation of the work.

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